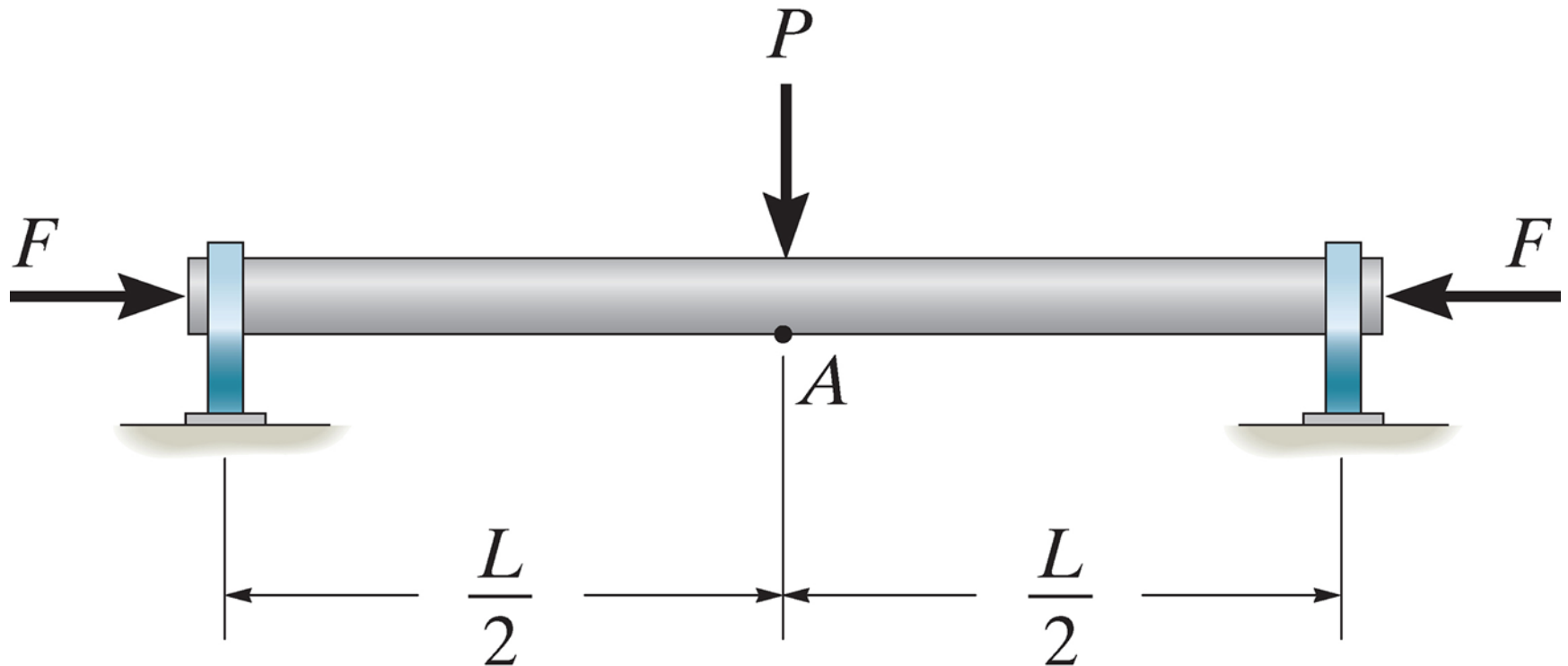


1. The shaft has a diameter $d = 20$ mm and is subjected to the loadings shown. If $P = 10$ kN and $F = 20$ kN, determine the principal stresses and the absolute maximum shear stress at the bottom surface point A . The total length of the shaft is $L = 1$ m. The bearings only support vertical reactions. (20' in total)



Solutions to Assignment #1:

Step 1: Support Reactions

Using Free-Body-Diagram, we get the support reactions:

- Total force in y -direction:

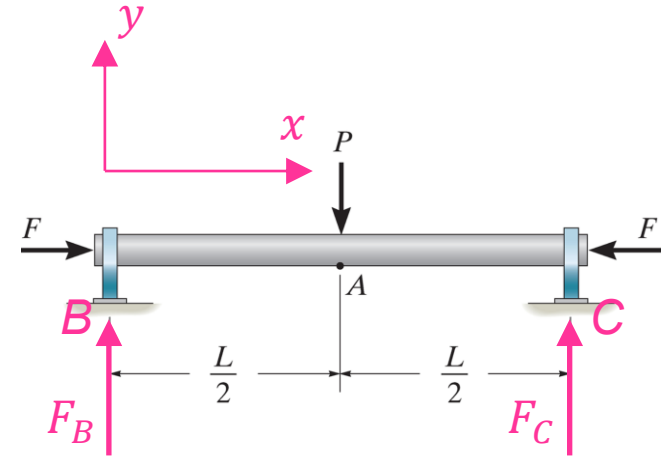
$$\sum F_y = F_B + F_C - P = 0 \quad (1)$$

- Total moment:

$$\sum M_A = F_B \times L/2 - F_C \times L/2 = 0 \quad (2)$$

Solving Eqs. (1) and (2), we have:

$$F_B = F_C = 5 \text{ kN}$$



Solutions to Assignment #1, cont'd:

Step 2: Internal Loadings

Taking a section at point A and using Free-Body-Diagram, we get the internal resultant forces and moment:

- Total force in y -direction:

$$\sum F_y = F_B - (V + P) = 0$$

$$\rightarrow \text{Shear force } V = -5 \text{ kN}$$

- Total force in x -direction:

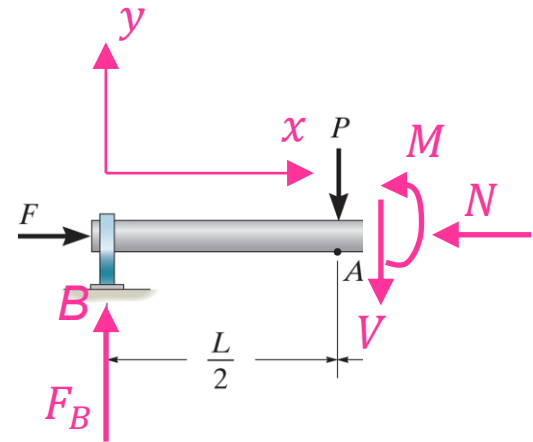
$$\sum F_x = F - N = 0$$

$$\rightarrow \text{Normal force } N = 20 \text{ kN (Compressive)}$$

- Total moment:

$$\sum M_A = M - F_B \times L/2 = 0$$

$$\rightarrow \text{Internal moment } M = 2.5 \text{ kN} \cdot \text{m}$$



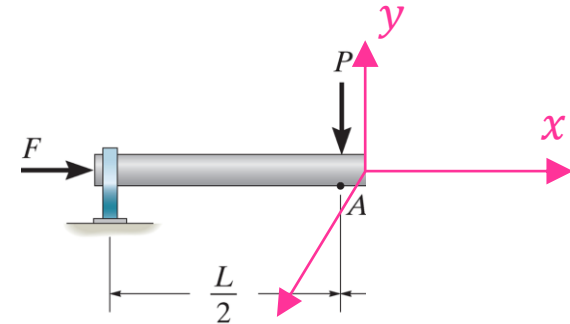
Solutions to Assignment #1, cont'd:

Step 3: Stress State at the Surface Point A

Both (compressive) force and moment will induce normal stress to the surface point.

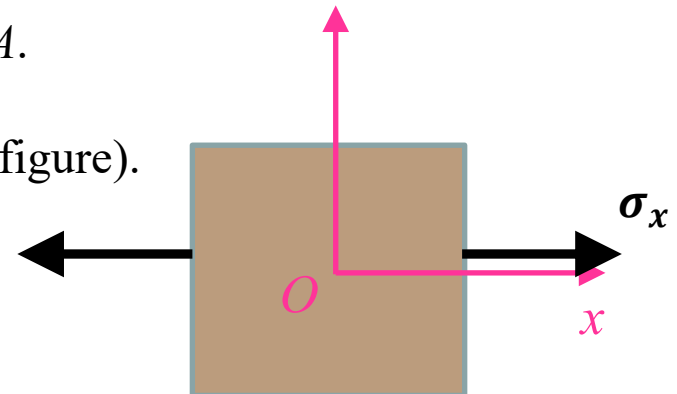
Establish $Oxyz$ coordinate system and the A point, with x -axis along the longitudinal direction of the shaft.

$$\begin{aligned}\sigma_x &= -\frac{N}{A} + \frac{My}{I} \\ &= -\frac{20000}{\frac{\pi(0.02^2)}{4}} + \frac{2500 \times \frac{0.02}{2}}{\frac{\pi(0.01^4)}{4}} \\ &= 3119.4 \text{ MPa}\end{aligned}$$



There is no shear stress induced at the surface point A .

The plane stress state in Oxy system is plotted (right figure).



Solutions to Assignment #1, cont'd:

Step 4: Construction of the Mohr's Circle

Given $\sigma_x = 3119.4 \text{ MPa}$, $\sigma_y = 0$, $\tau_{xy} = 0$:

Since there is no shear stress on the element, we can immediately know the principal stresses as:

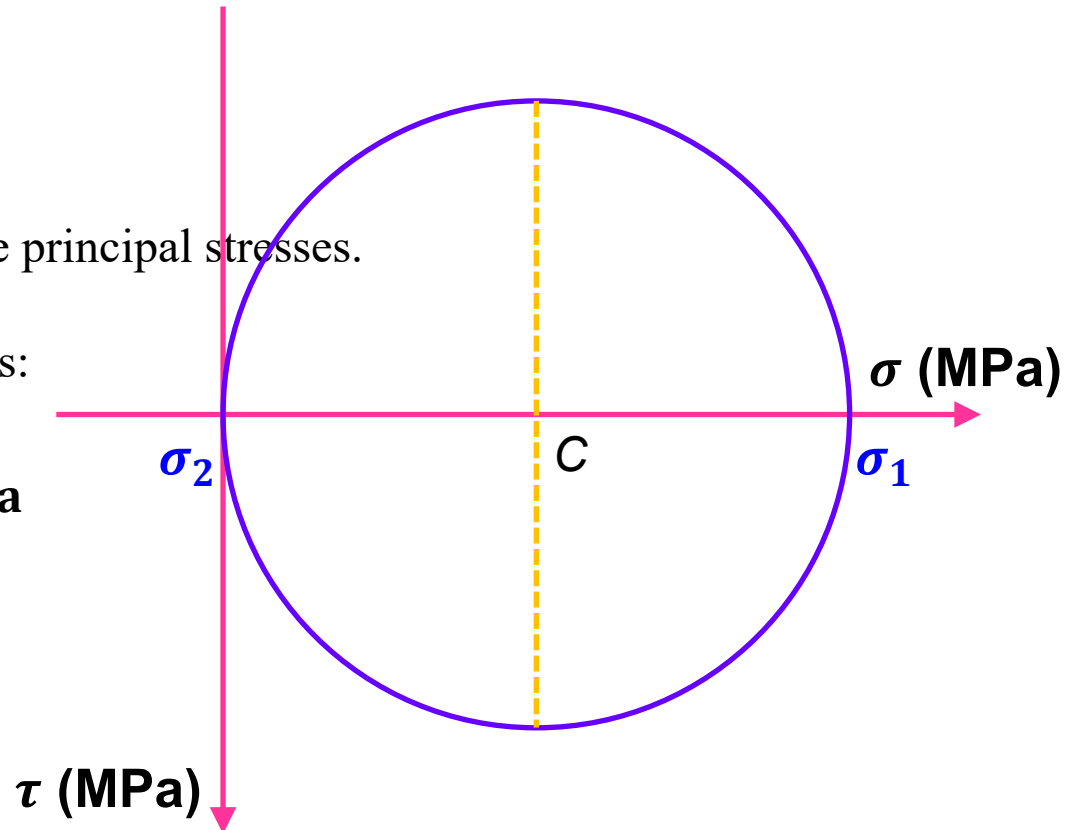
$$\sigma_1 = \sigma_x = \mathbf{3119.4 \text{ MPa}} \quad (\text{Ans.})$$

$$\sigma_2 = \mathbf{0} \quad (\text{Ans.})$$

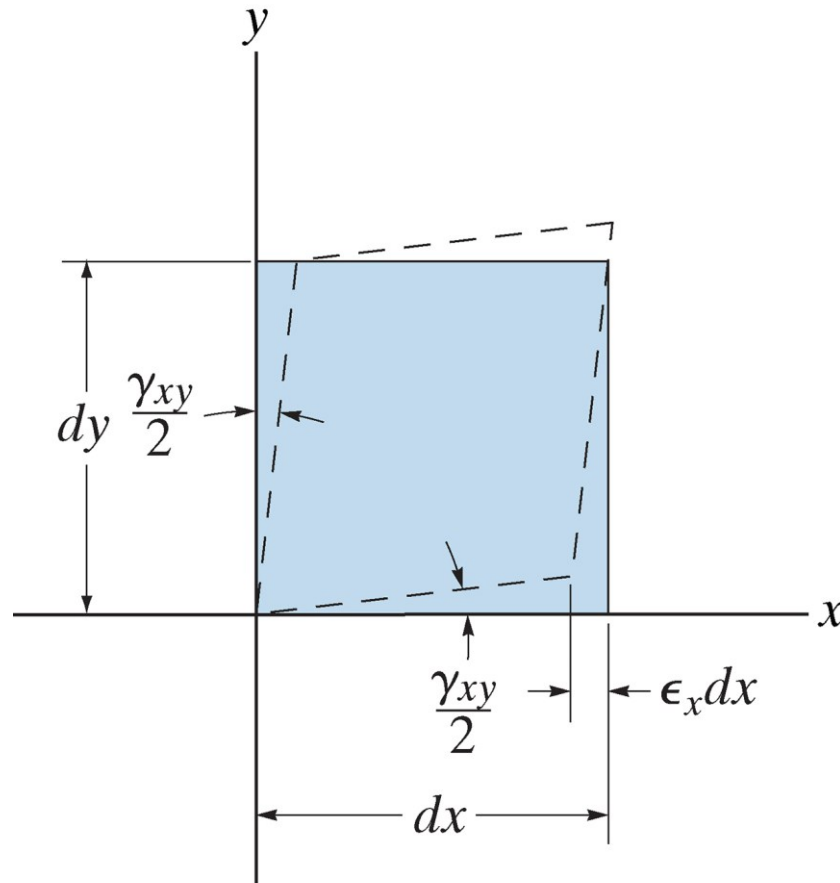
The circle is constructed based on the principal stresses.

The absolute maximum shear stress is:

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1 - \sigma_2}{2} = R = \mathbf{1559.7 \text{ MPa}} \quad (\text{Ans.})$$



2. The state of plane strain at a point has strain components of $\epsilon_x = -200 \times 10^{-6}$, $\epsilon_y = 0$, and $\gamma_{xy} = 100 \times 10^{-6}$. Determine (a) the principal strains; (b) the maximum in-plane shear strain and the corresponding orientation of the principal plane; and (c) the absolute maximum shear strain at the point. (15' points)



Solutions to Assignment #2:

This is a typical plane strain problem. Given $\varepsilon_x = -200 \times 10^{-6}$, $\varepsilon_y = 0$, $\gamma_{xy}/2 = 50 \times 10^{-6}$, the Mohr's circle for the plane strain can be constructed.

$$\text{Center } C\left(\frac{\varepsilon_x + \varepsilon_y}{2}, 0\right) \rightarrow C(-100 \times 10^{-6}, 0)$$

$$\text{Reference point } A(-200 \times 10^{-6}, 50 \times 10^{-6})$$

$$\text{Radius: } R = \sqrt{(-100 + 200)^2 + 50^2} \times 10^{-6} = 111.8 \times 10^{-6}$$

Principal strains:

$$\varepsilon_1 = (-100 + 111.8) \times 10^{-6} = 11.8 \times 10^{-6}$$

$$\varepsilon_2 = (-100 - 111.8) \times 10^{-6} = -211.8 \times 10^{-6}$$

Orientation:

$$\text{Counterclockwise } \theta = 76.7^\circ \text{ (corresponding to } \varepsilon_1)$$

Maximum in-plane shear strain:

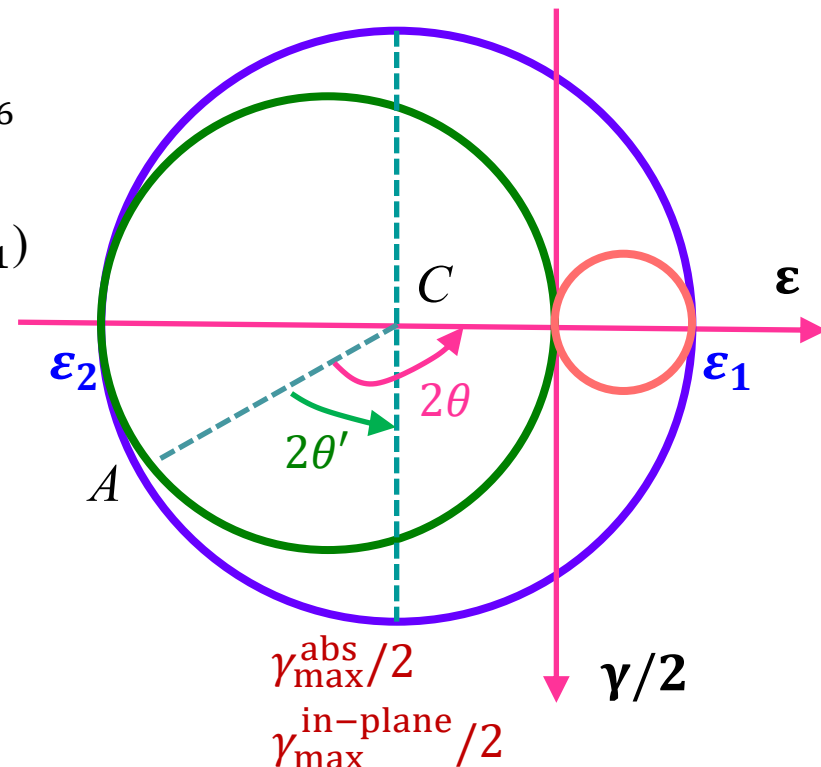
$$\gamma_{\max}^{\text{in-plane}} = 2R = \mathbf{223.6 \times 10^{-6}} \quad (\text{Ans.})$$

Orientation:

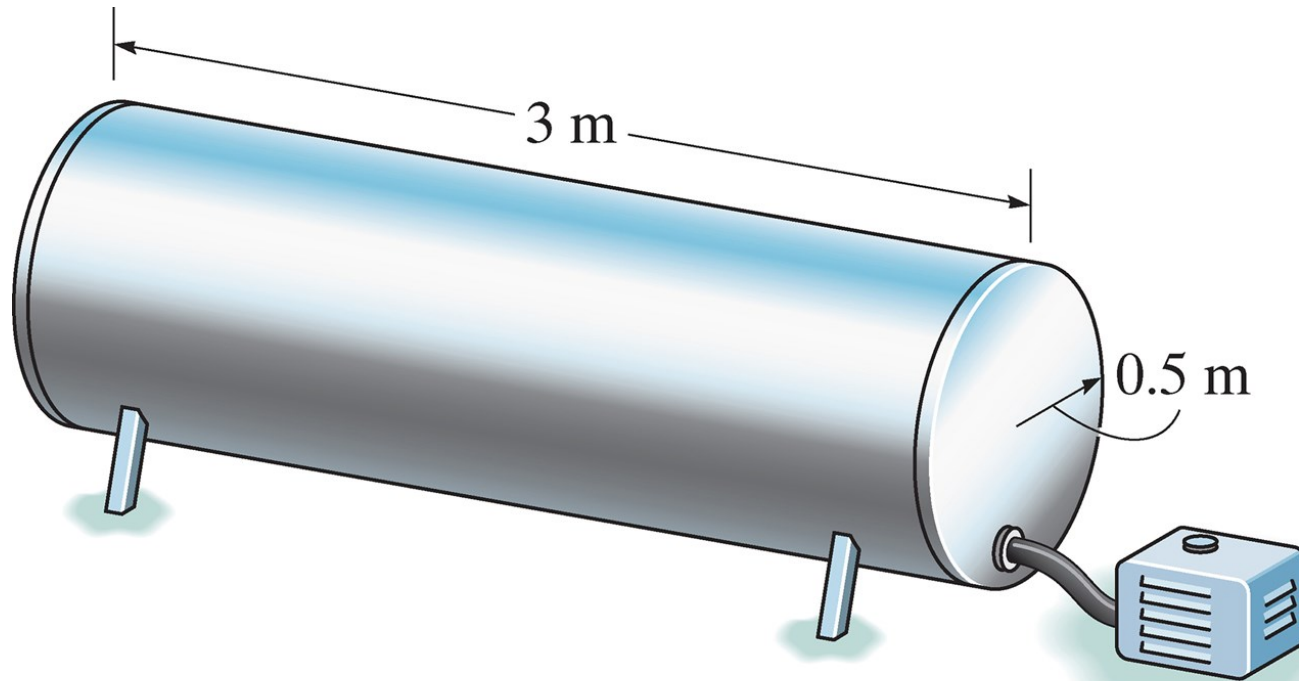
$$\text{Counterclockwise } \theta' = 31.7^\circ$$

The absolute maximum shear strain:

$$\gamma_{\max}^{\text{abs}} = 2R = \mathbf{223.6 \times 10^{-6}} \quad (\text{Ans.})$$



3. The thin-walled cylindrical pressure vessel of inner radius r and thickness t is subjected to an internal pressure p . Calculate the increase in both the diameter and the length of a steel pressure vessel filled with air and having an internal gas pressure of 15 MPa. The vessel is 3 m long and has an inner radius of 0.5 m and a thickness of 10 mm. Take $E = 200$ GPa, $\nu = 0.3$. (15' in total)



Solutions to Assignment #3:

The state of stress for any point located on the vessel's wall can be represented by:

$$\sigma_1 = \frac{pr}{t} = \frac{15 \times 10^6 \times 0.5}{0.01} = 7.5 \times 10^8 \text{ Pa} = 750 \text{ MPa}$$

$$\sigma_2 = \frac{pr}{2t} = \frac{15 \times 10^6 \times 0.5}{2 \times 0.01} = 3.75 \times 10^8 \text{ Pa} = 375 \text{ MPa}$$

According the generalized Hooke's law, the strain in the longitudinal direction is:

$$\varepsilon_{\text{long}} = \frac{1}{E} [\sigma_2 - \nu \sigma_1] = \frac{3.75 \times 10^8 - 0.3 \times 7.5 \times 10^8}{200 \times 10^9} = 0.75 \times 10^{-3}$$

The strain in the hoop direction is:

$$\varepsilon_{\text{hoop}} = \frac{1}{E} [\sigma_1 - \nu \sigma_2] = \frac{7.5 \times 10^8 - 0.3 \times 3.75 \times 10^8}{200 \times 10^9} = 3.1875 \times 10^{-3}$$

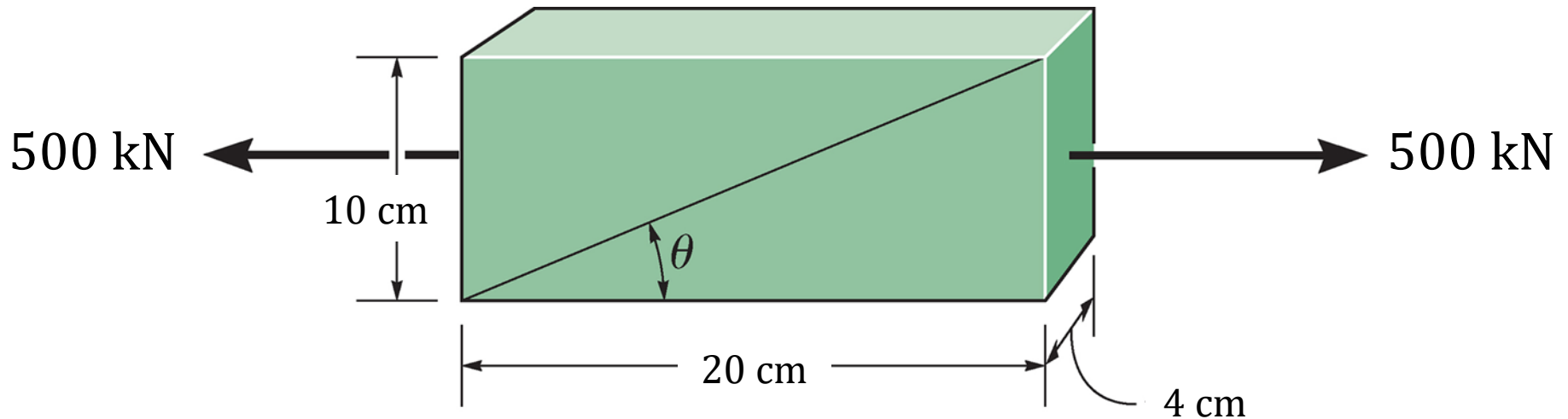
→ The increase in the length:

$$\Delta L = \varepsilon_{\text{long}} L_0 = 0.75 \times 10^{-3} \times 3 = 2.25 \times 10^{-3} \text{ m} = \mathbf{2.25 \text{ mm}} \quad (\text{Ans.})$$

The increase in the diameter:

$$\Delta D = \varepsilon_{\text{hoop}} D_0 = 3.1875 \times 10^{-3} \times 1 = 3.1875 \times 10^{-3} \text{ m} = \mathbf{3.1875 \text{ mm}} \quad (\text{Ans.})$$

4. The steel bar is subjected to an axial force of 500 kN. If it has the original dimensions shown, determine the change in the angle θ after the load is applied. Take $E = 200$ GPa and $\nu = 0.3$. (10' points)



Solutions to Assignment #4:

The steel bar is subjected to uniaxial loading, i.e.

$$\sigma_x = \frac{500 \times 10^3}{0.04 \times 0.1} = 125 \times 10^6 \text{ Pa}$$

$$\sigma_y = 0$$

$$\sigma_z = 0$$

Applying the general Hooke's law we have

$$\varepsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)] = \frac{125 \times 10^6}{200 \times 10^9} = 6.25 \times 10^{-4}$$

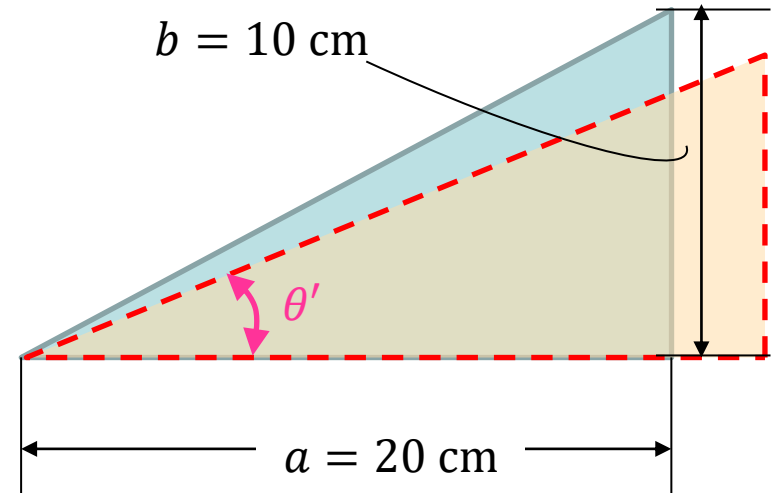
$$\varepsilon_y = \frac{1}{E} [\sigma_y - \nu(\sigma_x + \sigma_z)] = -0.3 \times \frac{125 \times 10^6}{200 \times 10^9} = -1.875 \times 10^{-4}$$

$$\text{New length } a' = a(1 + \varepsilon_x) = 0.2 \times (1 + 6.25 \times 10^{-4}) = 0.200125 \text{ m}$$

$$\text{New high } b' = b(1 + \varepsilon_y) = 0.1 \times (1 - 1.875 \times 10^{-4}) = 0.09998125 \text{ m}$$

$$\text{New angle } \theta' = \arctan \frac{b'}{a'} = 26.546^\circ$$

$$\text{Original angle } \theta = \arctan \frac{b}{a} = 26.565^\circ$$

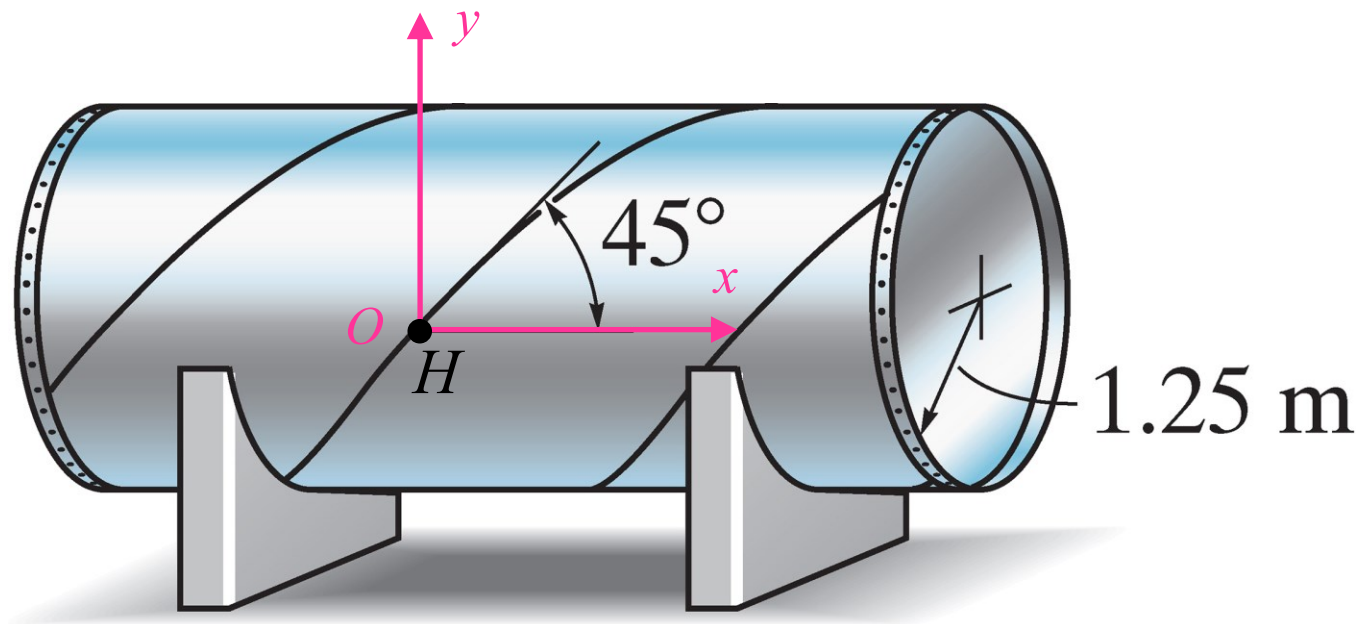


The change in the angel $\Delta\theta = -0.019^\circ$

(Ans.)

The negative sign indicates the decrease in the angle.

5. The cylindrical pressure vessel has an inner radius of 1.25 m and a wall thickness of 15 mm. Determine the circumferential (hoop) and longitudinal strains and the absolute maximum shear strain for a point H on the surface if the vessel is subjected to an internal pressure of 8 MPa. The steel has modulus of elasticity $E = 200$ GPa and Poisson's ratio $\nu = 0.30$. (20' in total)



Solutions to Assignment #5:

In gas filled vessel, for the Oxy coordinate system established as shown, there are non-equal normal stresses in two perpendicular directions (one in the longitudinal direction and the other in the hoop direction).

Therefore, the principal stresses are $\sigma_1 = \sigma_{\text{hoop}}$, $\sigma_2 = \sigma_{\text{long}}$. Note that $\sigma_{\text{hoop}} = 2\sigma_{\text{long}}$, thus $\sigma_1 > \sigma_2 > 0$.

$$\sigma_{\text{long}} = \frac{pr}{2t} = \frac{8 \times 10^6 \times 1.25}{2 \times 0.015} = 333.3 \text{ MPa}$$

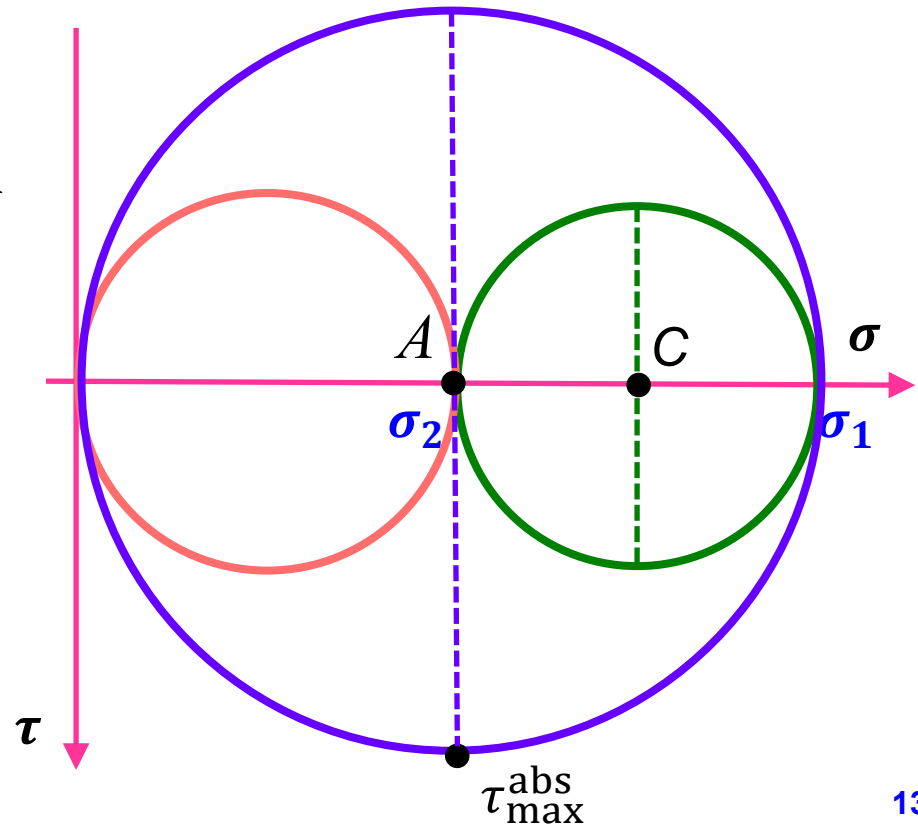
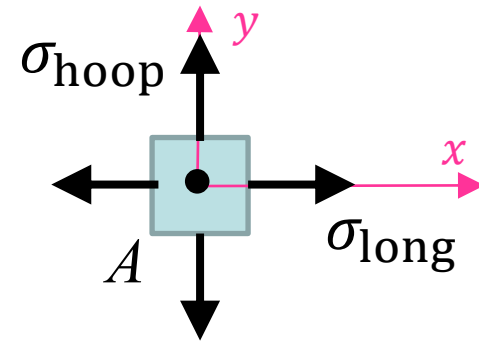
$$\sigma_{\text{hoop}} = 2\sigma_{\text{long}} = 666.6 \text{ MPa}$$

$$R = 166.7 \text{ MPa}$$

The reference point A is also shown on the Mohr's circle.

The absolute maximum shear stress is

$$\tau_{\text{max}}^{\text{abs}} = \frac{\sigma_1}{2} = \frac{\sigma_a}{2} = 333.3 \text{ MPa}$$



Solutions to Assignment #5, cont'd:

With $\sigma_x = \sigma_n = 333.3 \text{ MPa}$, $\sigma_y = \sigma_a = 666.6 \text{ MPa}$, $\sigma_z = 0$, using generalized Hooke's law, we get:

The circumferential (hoop) strain (in y-direction)

$$\epsilon_{\text{hoop}} = \frac{\sigma_{\text{hoop}} - \nu \sigma_{\text{long}}}{E} = \frac{666.6 \times 10^6 - 0.3 \times 333.3 \times 10^6}{200 \times 10^9} = \mathbf{2.8331 \times 10^{-3}} \quad (\text{Ans.})$$

and longitudinal strain (in x-direction)

$$\epsilon_{\text{long}} = \frac{\sigma_{\text{long}} - \nu \sigma_{\text{hoop}}}{E} = \frac{333.3 \times 10^6 - 0.3 \times 666.6 \times 10^6}{200 \times 10^9} = \mathbf{6.666 \times 10^{-4}} \quad (\text{Ans.})$$

The shear modulus can be obtained by the material property relationship

$$G = \frac{E}{2(1 + \nu)} = \frac{200 \times 10^9}{2 \times (1 + 0.3)} = 76.923 \text{ GPa}$$

Finally, the absolute maximum shear strain is obtained by using the maximum shear stress and Hooke's law:

$$\gamma_{\text{max}}^{\text{abs}} = \frac{\tau_{\text{max}}^{\text{abs}}}{G} = \frac{333.3 \times 10^6}{76.923 \times 10^9} = \mathbf{4.333 \times 10^{-3} \text{ rad}} \quad (\text{Ans.})$$

6. If the strain at a point in a material has components of $\varepsilon_x = 50 \times 10^{-6}$, $\varepsilon_y = 100 \times 10^{-6}$, and $\gamma_{xy} = -120 \times 10^{-6}$, determine the principal **stresses** at the point. Assume the material has $E = 200$ GPa and $\nu = 0.3$, and the point is subjected to plane stress (**not plane strain**). (**20'** in total)

Solutions to Assignment #6:

Given $\varepsilon_x = 50 \times 10^{-6}$, $\varepsilon_y = 100 \times 10^{-6}$, and $\gamma_{xy} = -120 \times 10^{-6}$, applying Hooke's law in the xy -plane (plane stress in xy -plane, i.e. $\sigma_z = 0$), we have

$$\varepsilon_x = \frac{1}{E} [\sigma_x - \nu \cdot (\sigma_y + \sigma_z)] = \frac{\sigma_x - \nu \cdot \sigma_y}{E}$$

$$50 \times 10^{-6} = \frac{\sigma_x - 0.3 \times \sigma_y}{200 \times 10^9} \quad (1)$$

$$\varepsilon_y = \frac{1}{E} [\sigma_y - \nu \cdot (\sigma_x + \sigma_z)] = \frac{\sigma_y - \nu \cdot \sigma_x}{E}$$

$$100 \times 10^{-6} = \frac{\sigma_y - 0.3 \times \sigma_x}{200 \times 10^9} \quad (2)$$

Solving Eqs. (1) and (2) yields

$$\sigma_x = 17.6 \text{ MPa}$$

$$\sigma_y = 25.3 \text{ MPa}$$

Solutions to Assignment #6, cont'd:

The shear stress can be determined using the Hooke's law for shear. First, we calculate shear modulus by

$$G = \frac{E}{2(1 + \nu)} = \frac{200}{2 \times (1 + 0.3)} = 76.9 \text{ GPa}$$

Thus,

$$\tau_{xy} = G\gamma_{xy} = 76.9 \times 10^9 \times (-120 \times 10^{-6}) = -9.23 \text{ MPa}$$

Combining σ_x , σ_y , and τ_{xy} we can draw the Mohr's circle of plane stress.

Center $C(21.45, 0)$

Reference point $A(17.6, -9.23)$

$$\begin{aligned} \text{Radius } R &= \sqrt{(21.45 - 17.6)^2 + 9.23^2} \\ &= 10 \text{ MPa} \end{aligned}$$

The principal stresses are

$$\sigma_1 = 21.45 + 10 = \mathbf{31.45 \text{ MPa}} \quad (\text{Ans.})$$

$$\sigma_2 = 21.45 - 10 = \mathbf{11.45 \text{ MPa}} \quad (\text{Ans.})$$

